

Semiconductor manufacturers like TI and STMicroelectronics offer wireless transceivers, microcontrollers, development platforms and more. Test and measurement leaders like National Instruments have a variety of design, testing and development tools for this space. Solar specialists like Racold Thermo are also innovating at a rapid pace to cater to the rising demand for renewable energy options.

It is obvious that there is endless innovation and new product development happening in this space. Although the extent of adoption of such technologies is comparatively low in India, it is slowly but surely on the rise here too. Sitting on the tip of the iceberg, this is probably the ideal time for us to delve into some of the prominent technological trends in the energy sector.

## Smarter energy management

One of the changes that have taken over the energy sector is the use of energy management systems by large residential complexes, hospitals, offices, hotels, industries and even individual homes.

"Technological innovations in processors, networking and wireless technologies are enabling this sector to make tremendous strides in making energy management smarter," says Satish Mohanram, technical marketing manager, National Instruments India. He cites some examples:

**Smart metering.** Fitted at the consumer end, these meters can measure and control loads based on the availability of power. It also helps in controlling power thefts.

**Smart networks.** Wide-area networks with high-speed measurement capabilities like phasor measurement help predict grid instability and hence avoid blackouts.

**Data mining.** High-end servers perform data mining to understand consumption patterns to effectively control generation and load scheduling. This also helps in minimising transmission losses by optimising transmission routes.



Wiser for home automation by Schneider Electric

**More power.** Integration of renewable energy sources, better battery technology to store energy, and efficient inverters help make up for the shortage of generated power.

**Predictive maintenance.** Smart switches, condition monitoring systems, etc, increase the life of power generation and transmission lines through predictive maintenance.

**Smart chiller management systems.** These save energy by operating during off peak hours and reducing consumption during peak industrial demand.

In India, due to the shortage of power, most of these innovations are very relevant. There is a decent level of awareness about the same and hence these innovations are catching up. "The Smart Grid implementation as a part of the R-APDRP project is a big step in India in this direction," says Mohanram.

## Smart grids as basis of energy revolution

When we speak of new technologies in the energy space, it is the 'smart grid' that weaves it all together. A smart grid is like a backbone that connects the entire energy value chain, right from the sources up to the consumers. Efficient management of the grid can deliver maximum value to users from the resources tapped.

Sapna Mongia, technical marketing manager-South Asia operations, STMi-

croelectronics, says, "Governments and power companies across the world have recognised that the traditional power grid, which has not changed significantly in 100 years, must be replaced with more efficient, flexible and intelligent energy-distribution networks called smart grids. With 17 per cent of the world's population residing in India, the multi-fold increase in the demand of electric supply makes it all the more essential to have a successful smart grid running across the country."

Smart grid technology enables consolidation of varied power sources, end-user energy management, minimisation of power disruptions, and transportation of only the required amount of power. All this results in more balanced usage, better matching of supply to demand, lower cost to the customer, more reliable power supply and reduced carbon emissions.

"The government of India, private companies, utilities, distribution companies and startups are in close collaboration with the government and engaged in many smart grid projects," explains Dr Kumar. "The government has set up the Task Force on Smart Grids, then there is the India Smart Grid Forum (ISGF). IEEMA also has a task force for smart grids. We are members of the ISGF and IEEMA's task force on smart grids."

Schneider Electric is involved in the grid optimisation layer of the Indian smart grid market. Grid optimisation serves as an umbrella term that encompasses improvement in three main areas, namely, system reliability, operational efficiency, and asset utilisation and protection.

## Smart metering

Smart meters are intelligent devices that connect every consumption point to the smart grid, enabling controlled supply of power and personalised services.

"Smart metering essentially involves an electronic power meter supplemented by full remote control, diagnostics, power peak and consumption analysis, anti-tampering mechanisms, fault alerts, time-variable tariffs